

# Meeting Timer

QUICK USER GUIDE | ECE 463

**Presets: 15, 20, 25 and 30 minutes.** The current program has no 10-minute preset.

**SET:** closer to the XIAO (D4/GPIO5). **RUN:** farther away (D5/GPIO6).

These positions describe the original layout with the XIAO on the left. If moved, use labels/pins. A tap is press-and-release; a hold is about 1.5 seconds.

## 1. Power on and set the hand to zero

1. Connect USB with both buttons released. If needed, start the current timer program in Thonny with **F5**.
2. While the LED pulses **purple**, tap **SET** to move forward (hold to keep moving), or tap **RUN** to move backward.
3. Align the hand with **0/30**, release the buttons and wait for it to stop. Hold **RUN for 1.5 seconds**, then release to confirm zero and enter selection.

## 2. Select the time, then start

After a **fresh power-up and zero calibration**, 15 minutes is selected. Counts below start there.

| Duration   | Extra SET taps | Blue pulses/group | Start after hand stops |
|------------|----------------|-------------------|------------------------|
| 15 minutes | 0              | 1                 | Tap RUN once           |
| 20 minutes | 1              | 2                 | Tap RUN once           |
| 25 minutes | 2              | 3                 | Tap RUN once           |
| 30 minutes | 3              | 4                 | Tap RUN once           |

**Wait for the hand to stop before RUN.** Initial 15-minute positioning takes about 27 seconds. SET cycles **15 > 20 > 25 > 30 > 15**. Later, count blue pulses to identify the selection; groups have a two-second gap. SET selects only before starting or after completion.

## 3. Read the LED

| LED indication                        | Meaning                                |
|---------------------------------------|----------------------------------------|
| <b>Purple pulses</b>                  | Set the hand's zero position           |
| <b>1/2/3/4 blue pulses + gap</b>      | Selecting 15/20/25/30 minutes          |
| <b>Green pulses</b>                   | Running; more than 5 minutes remain    |
| <b>Red pulses while counting down</b> | Running; 5 minutes or less remain      |
| <b>Repeating blue, no group gap</b>   | Paused                                 |
| <b>Red pulses, hand at zero</b>       | Finished; continues until acknowledged |

Each brightness pulse lasts one second. There is no buzzer. **0 and 30 share one dial mark**; use the LED state to distinguish them.

## 4. Pause, restart or reset

**Pause/resume:** tap RUN to pause; once settled, tap RUN again to resume.

**Cancel/reload:** hold RUN for 1.5 seconds. Wait for the hand to position, then tap RUN to restart.

**After completion:** tap RUN to reload; wait, then tap RUN again to start. Or tap SET for the next preset.

**Recalibrate:** outside purple mode, hold SET for 1.5 seconds and repeat zero-setting. The selected preset is retained.

**Turn off:** unplug USB. Every power-up/reset needs zero calibration; an interrupted countdown is lost.

Checked 2026-09-15 against code/meeting\_timer.py, QUICK\_TEST = False. Accelerated test mode uses shorter durations.